

A Comparison of Test Statistics for the Equality of Covariance Matrices in High-Dimensional Data and an Application to Microarray Data

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Abstract

The purpose of this study is to compare the performance of five statistical tests: the test statistic T_1 by Schott (2007), the test statistic T_2 by Srivastava and Yanagihara (2010) the test statistic T_3 by Ahmad (2017) the test statistic T_4 by Qayed and Han (2020), and the test statistic T_5 by Sun et al. (2022), for testing the equality of covariance matrices in high-dimensional data with a multivariate normal distribution. The comparison was conducted using Monte Carlo simulation techniques under five matrix structures: First-Order Autoregressive (AR1), Compound Symmetry, Simple, Banded Toeplitz, and Variance Components. The criteria used were the ability to control the type I error rate and power. The results showed that the test statistics T_3 T_4 and T_5 were able to control the type I error rate I error, while T_1 and T_2 could not. When applied to gene expression data from glioma brain tumors, the test statistic T_3 provided the most reliable results, as it exhibited the highest power.

Keywords: Multivariate Normal Distribution, High-Dimensional Data, Type I error rate, Power of a test.

Introduction

In multivariate analysis, testing the equality of covariance matrices between two populations is commonly conducted using Box's Test and the Likelihood Ratio Test (LRT). These tests are applicable when the number of variables p is smaller than the sample sizes of both groups n_1 and n_2 . However, when the number of variables is greater than or equal to the sample size, this situation is referred to as high-dimensional data. Box's Test and the LRT are no longer applicable because the sample covariance matrix cannot be inverted.

To address this issue, many statisticians have developed new test statistics specifically designed for covariance matrix testing in high-dimensional settings. These methods focus on properly controlling the type I error rate while maintaining adequate power. Previous studies have shown that the test statistics proposed by Schott (2007), as well as those by Srivastava and Yanagihara (2010), have received considerable attention and have been widely applied in this context. Subsequently, additional test statistics have been developed to overcome various limitations associated with high-dimensional data. Ahmad (2017) and Qayed and Han (2020) proposed testing procedures that remain valid under non-normal distributions and diverse data conditions. Meanwhile, Sun et al. (2022) introduced a test statistic that demonstrated strong performance based on simulation studies. Therefore, these five test statistics were selected for comparison in the present study because they are well recognized in the literature and collectively cover a range of high-dimensional scenarios.

Accordingly, this study focuses on comparing test statistics for the equality of covariance matrices across multiple populations in high-dimensional settings. The five test statistics considered are those proposed by Schott (2007), Srivastava and Yanagihara (2010), Ahmad (2017), Qayed and Han (2020), and Sun et al. (2022). The comparison is conducted under different covariance matrix structures by evaluating the ability of each test to control the type I error rate and its power. The findings are expected to provide useful guidance for selecting appropriate test statistics in practical high-dimensional data analysis.

Materials and Methods

This study used desktop computers and laptops for data management, processing, and analysis. Python 3.10 was used in conjunction with Google Colab and Microsoft Excel for computation, data analysis, and result presentation. Internet connectivity was also employed to access databases and run the required programs.

This study focuses on comparing test statistics for the equality of covariance matrices across multiple populations in situations where the number of variables is greater than or equal to the sample size. The comparison is conducted using Monte Carlo simulation. Five test statistics are examined: the tests proposed by Schott (2007) (T_1), Srivastava and Yanagihara (2010) (T_2), Ahmad (2017) (T_3), Qayed and Han (2020) (T_4), and Sun et al. (2022) (T_5).

The study is conducted under the following scope:

1. The study assumes a multivariate normal distribution.
2. The study considers $k = 3$ and $k = 4$ populations.
3. The study considers $p \in \{25, 50, 100, 150, 300\}$ variables.
4. Five covariance matrix structures are considered, as follows:

4.1 First-Order Autoregressive (AR1) structure the hypotheses are defined as:

$$H_0^1: \Sigma_1 = \dots = \Sigma_k = A_0 \text{ vs } H_1^1: \Sigma_1 = \dots = \Sigma_{k-1} = A_0 \text{ where } \Sigma_k = A_1$$

where $A_0 = (\sigma_{ij})_{p \times p} = \begin{cases} 1 & ; |i-j|=0 \\ 0.7^{|i-j|} & ; |i-j| \neq 0 \end{cases}$ and $A_1 = (\sigma_{ij})_{p \times p} = \begin{cases} 1 & ; |i-j|=0 \\ 0.9^{|i-j|} & ; |i-j| \neq 0 \end{cases}$

4.2 Compound Symmetry (CS) structure the hypotheses are defined as:

$$H_0^2: \Sigma_1 = \dots = \Sigma_k = C_0 \text{ vs } H_1^2: \Sigma_1 = \dots = \Sigma_{k-1} = C_0 \text{ where } \Sigma_k = C_1$$

where $C_0 = 0.99I_p + 0.011\mathbf{1}_p\mathbf{1}_p'$ and $C_1 = 0.90I_p + 0.11\mathbf{1}_p\mathbf{1}_p'$

4.3 Simple (SIM) structure the hypotheses are defined as:

$$H_0^3: \Sigma_1 = \dots = \Sigma_k = I_p \text{ vs } H_1^3: \Sigma_1 = \dots = \Sigma_{k-1} = I_p \text{ where } \Sigma_k = U_1$$

where I_p is the $p \times p$ identity matrix and $U_1 = \text{diag}(\sigma_{11}, \sigma_{12}, \dots, \sigma_{pp})$
with $\sigma_{ij} \sim U(2, 3), i = 1, 2, \dots, p$

4.4 Banded Toeplitz (BTOEP) structure the hypotheses are defined as:

$$H_0^4: \Sigma_1 = \dots = \Sigma_k = T_0 \text{ vs } H_1^4: \Sigma_1 = \dots = \Sigma_{k-1} = T_0 \text{ where } \Sigma_k = T_1$$

where $T_0 = (\sigma_{ij})_{p \times p} = \begin{cases} 1 & ; |i-j|=0 \\ -0.5 & ; |i-j|=1 \\ 0 & ; |i-j| \neq 0, 1 \end{cases}$ and $T_1 = (\sigma_{ij})_{p \times p} = \begin{cases} 1 & ; |i-j|=0 \\ -0.05 & ; |i-j|=1 \\ 0 & ; |i-j| \neq 0, 1 \end{cases}$

4.5 Variance Components (VC) structure the hypotheses are defined as:

$$H_0^5: \Sigma_1 = \dots = \Sigma_k = V_0 \text{ vs } H_1^5: \Sigma_1 = \dots = \Sigma_{k-1} = V_0 \text{ where } \Sigma_k = V_1$$

where $V_0 = \text{diag}(\sigma_{11}, \sigma_{12}, \dots, \sigma_{pp})$ with $\sigma_{ij} \sim U(1, 2), i = 1, 2, \dots, p$
and $V_1 = \text{diag}(\sigma_{11}, \sigma_{12}, \dots, \sigma_{pp})$ with $\sigma_{ij} \sim U(3, 4), i = 1, 2, \dots, p$

5. Sample sizes the data are set as follows:

$$(n', p) = (25, 25), (25, 50), (25, 100), (25, 150), (25, 300), (50, 50), (50, 100), \\ (50, 150), (50, 300), (100, 100), (100, 150), (100, 300), (150, 150), (150, 300), (300, 300), \text{ where } n_i = n_j = n'$$

6. Compute test statistics and critical values of five test statistics, T_1 by Schott (2007), the test statistic T_2 by Srivastava and Yanagihara (2010), the test statistic T_3 by Ahmad (2017), the test statistic T_4 by Qayed and Han (2020), and the test statistic T_5 by Sun et al. (2022). The null hypothesis H_0 is rejected when the computed test statistics T_1 , T_3 and T_5 exceed the critical value obtained from the standard normal

distribution table, or when computed test statistics T_2 and T_4 exceed the critical value obtained from the chi-square distribution table, at a significance level α of 0.05

7. Compute the empirical type I error rate, assuming that the population follows the H_0 : The Equality of Covariance Matrices by taking random sample data 5000 iterations and compute the empirical type I error rate ($\hat{\alpha}$) as follows :

$$\hat{\alpha} = \frac{\text{the number of times rejecting } H_0 \text{ when } H_0 \text{ is true}}{5,000}$$

Then take $\hat{\alpha}$ of each test statistic to compare with the criterion of Bardley (1978) at a significance level of 0.05, which has a value $\hat{\alpha}$ within [0.025 - 0.075]. The test statistics with the empirical type I error rate are in such criteria, Indicating that it is able to control the empirical type I error rate.

8. Compute the empirical power, assuming that the population follows the H_1 : At least one covariance matrix is different. Random sample data 5000 iterations and compute the empirical power as follows :

$$\text{empirical power} = \frac{\text{the number of times rejecting } H_0 \text{ when } H_0 \text{ is false}}{5,000}$$

If a test statistic has greater empirical power, it indicates that the test statistic is more efficient. Only the test statistics that control the type I error rate are used to compute the empirical power.

9. The test statistics that controlled the type I error rate were applied to gene expression data from glioma brain tumors (Glioma) obtained from Henry Ford Hospital. Data is available on website <https://www.ncbi.nlm.nih.gov/>. The code of the data set is GSE4290. This data set is an mRNA expression data set collected from brain tumor patients to improve the diagnosis of molecular glioma tumors with a total of 54,613 variables and a total sample unit of 180 samples, divided into 23 samples were from a group of non-tumor epilepsy patients (Non-Tumor, NT) and 157 samples were from tumor patients. The tumor group consisted of 26 astrocytoma (Astrocytoma, AT), 50 oligodendroglioma (Oligodendroglioma, ODG), and 81 glioblastoma (Glioblastoma, GBT) samples. variables with a coefficient of variation (CV) in the interval [0.25 - 0.5] were selected for analysis and conclusion.

Results and Discussion

The simulation study considered five illustrative covariance matrix structures. The results are presented as trend graphs of the type I error rate and the empirical power, as shown in Fig. 1 - Fig. 10. Figures 1 - 2 display the results for the First-Order Autoregressive (AR1) structure. Figures 3 - 4 present the results for the Compound Symmetry structure. Figures 5 - 6 illustrate the results for the Simple structure. Figures 7 - 8 show the results for the Banded Toeplitz structure, and Figures 9 - 10 present the results for the Variance Components structure.

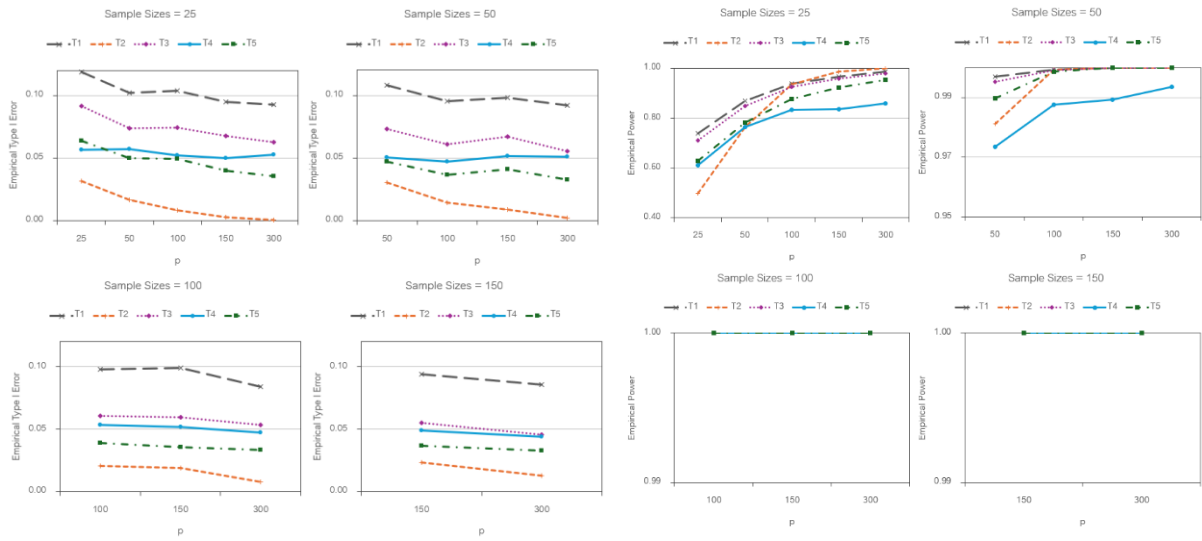


Fig. 1 Empirical Type I Error rate under H_0^1 and Empirical Power under H_1^1 in First-Order Autoregressive (AR1) structure when $k=3$

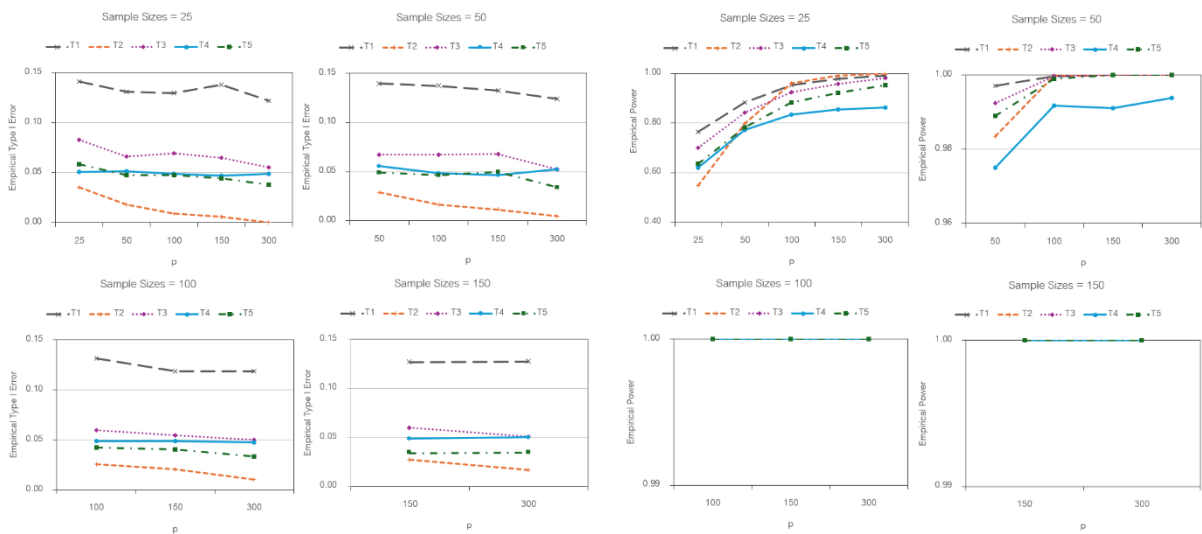


Fig. 2 Empirical Type I Error rate under H_0^1 and Empirical Power under H_1^1 in First-Order Autoregressive (AR1) structure when $k=4$

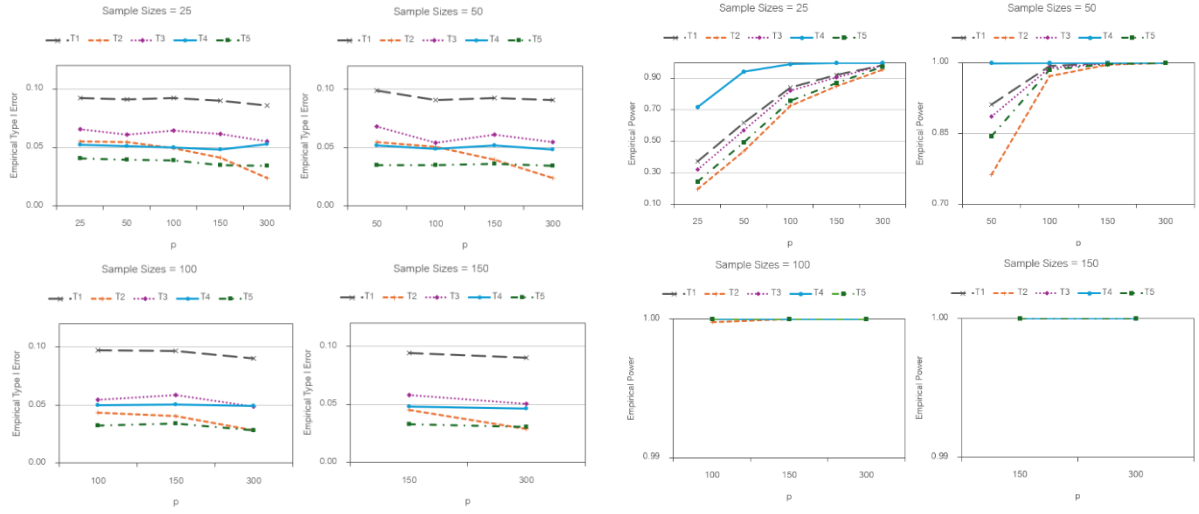


Fig. 3 Empirical Type I Error rate under H_0^2 and Empirical Power under H_1^2 in Compound Symmetry (CS) structure when $k=3$

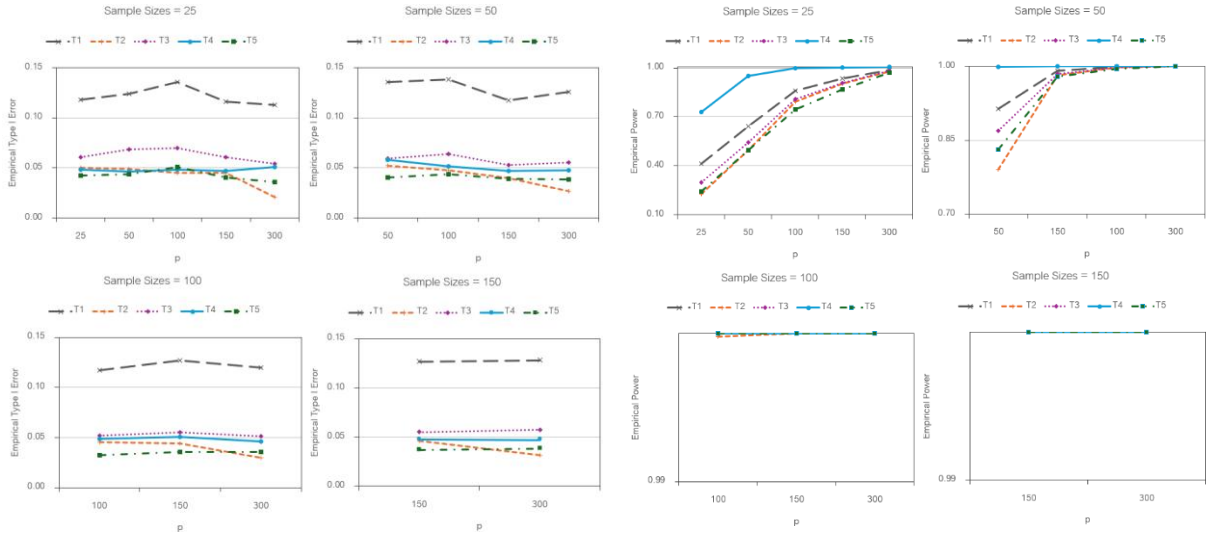


Fig. 4 Empirical Type I Error rate under H_0^2 and Empirical Power under H_1^2 in Compound Symmetry (CS) structure when $k=4$

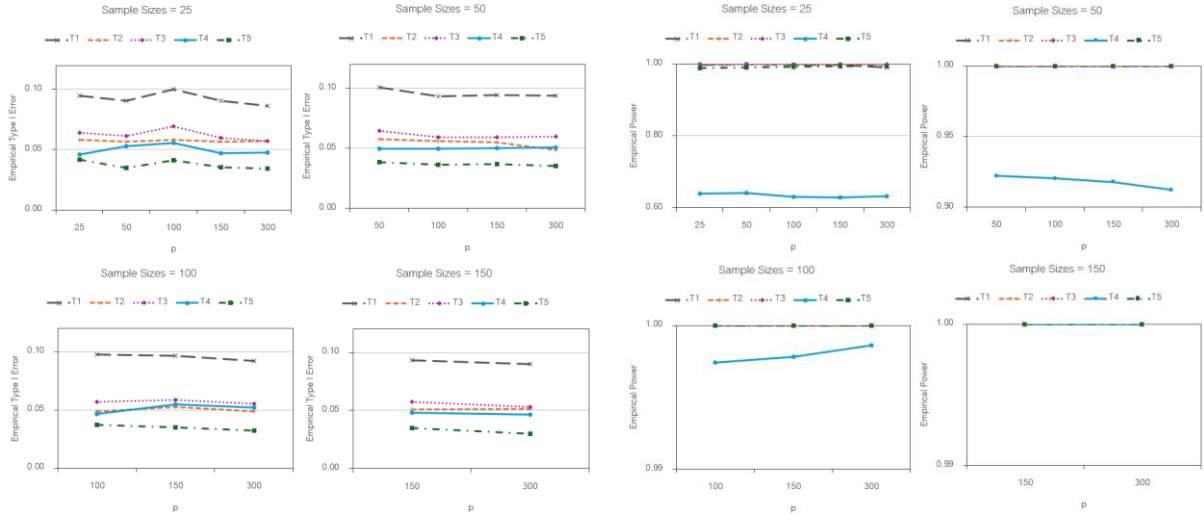


Fig. 5 Empirical Type I Error rate under H_0^3 and Empirical Power under H_1^3 in Simple structure when $k=3$

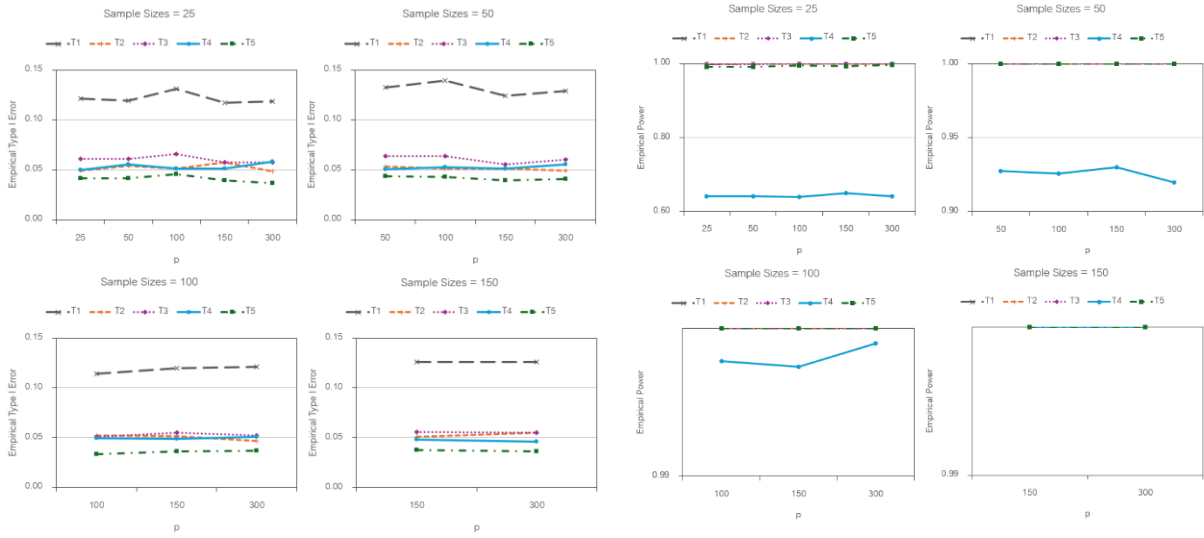


Fig. 6 Empirical Type I Error rate under H_0^3 and Empirical Power under H_1^3 in Simple structure when $k=4$

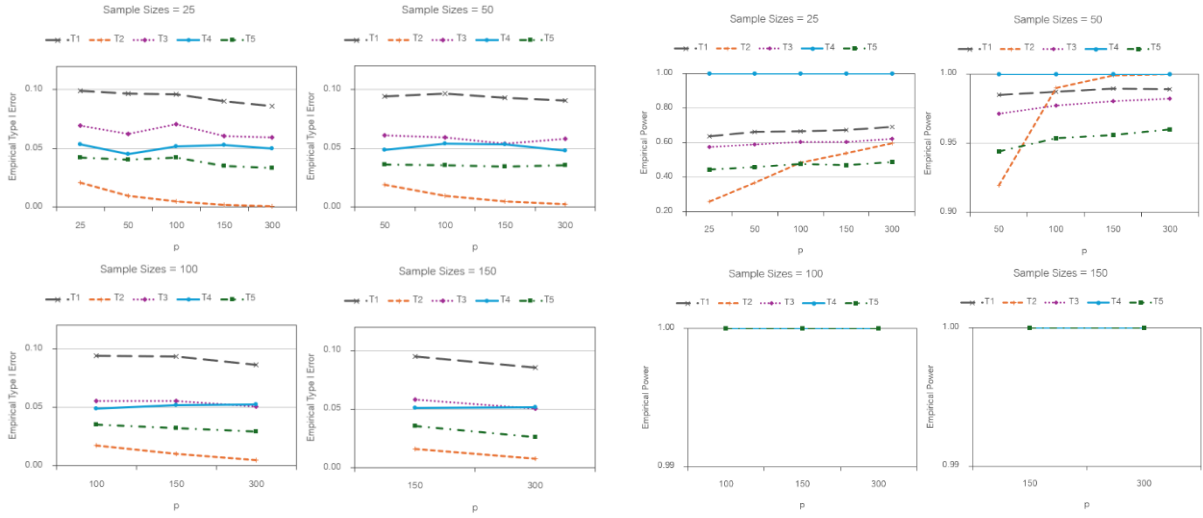


Fig. 7 Empirical Type I Error rate under H_0^4 and Empirical Power under H_1^4 in Banded Toeplitz (BTOEP) structure when $k=3$

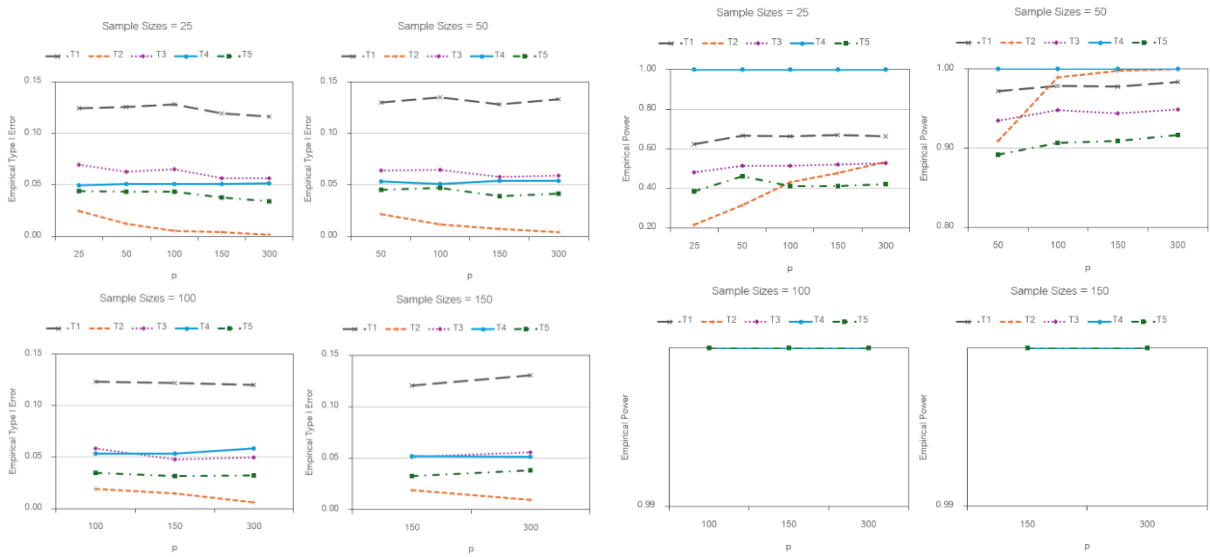


Fig. 8 Empirical Type I Error rate under H_0^4 and Empirical Power under H_1^4 in Banded Toeplitz (BTOEP) structure when $k=4$

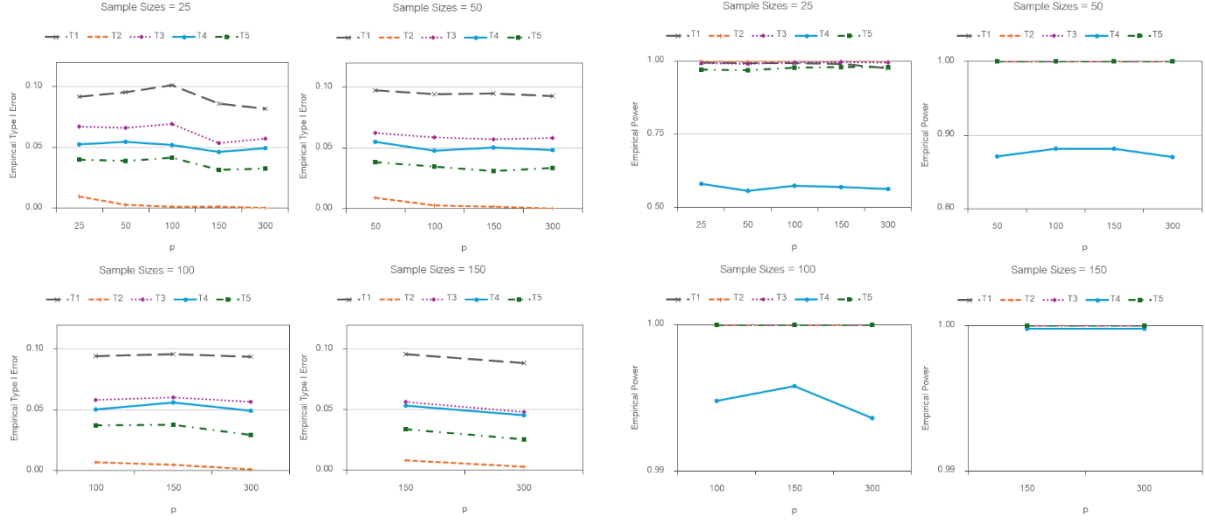


Fig. 9 Empirical Type I Error rate under H_0^5 and Empirical Power under H_1^5 in Variance Components (VC) structure when $k=3$

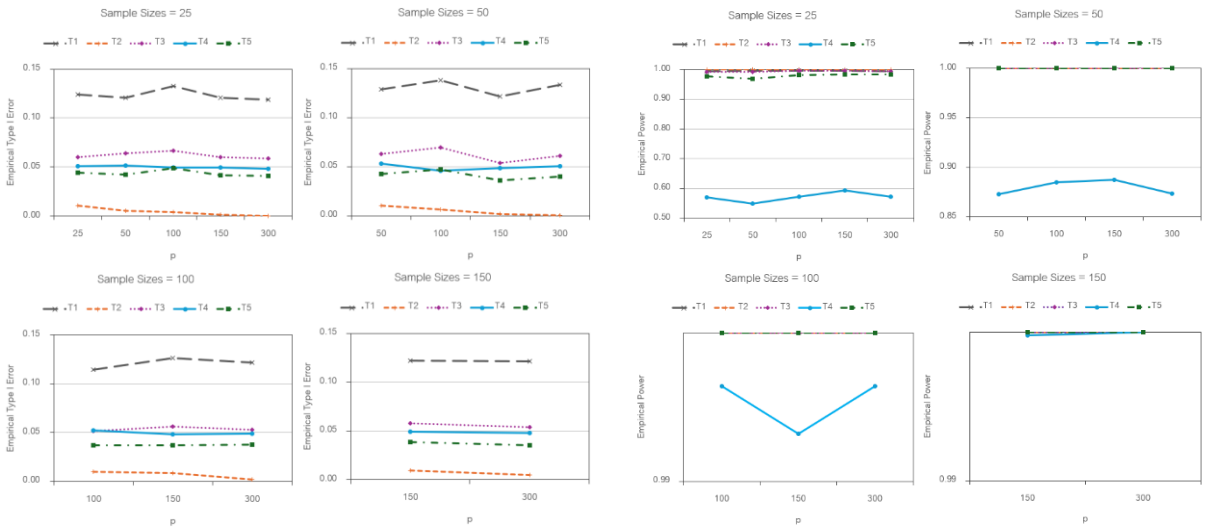


Fig. 10 Empirical Type I Error rate under H_0^5 and Empirical Power under H_1^5 in Variance Components (VC) structure when $k=4$

From Fig. 1 - Fig. 10, test statistic T_4 is able to control the empirical type I error rate within the criterion of Bardley's within the interval $[0.025-0.075]$, under the specified covariance structures and at a significance level of 0.05, particularly as the sample size and the number of variables increase. This holds for the First-Order Autoregressive (AR1), Compound Symmetry, Simple, Banded Toeplitz, and Variance Components structures. The empirical type I error rate of T_4 falls within $[0.025-0.075]$ in 100% of cases.

Similarly, test statistic T_3 is able to control the empirical type I error rate within $[0.025-0.075]$ in nearly 100% of cases across all covariance structures. Specifically, T_3 achieves 100% control under Compound Symmetry, Simple, Banded Toeplitz, and Variance Components structures. Under the First-Order

Autoregressive (AR1) structure, the empirical type I error rate falls within [0.025–0.075] in 93.33% of cases; however, it fails to meet the criterion when $(n', p) = (25, 25)$.

For test statistic T_5 the empirical type I error rate is controlled within [0.025–0.075] under all covariance structures in 100% of cases, similar to T_4 . In contrast, test statistic T_2 satisfies Bardley's (1978) criterion within [0.025–0.075] under only two covariance structures: Compound Symmetry and Simple. Under the First-Order Autoregressive (AR1) and Variance Components structures, T_2 fails to control the empirical type I error rate within the specified interval. Test statistic T_1 is unable to control the empirical type I error rate within Bardley's acceptable range [0.025–0.075] under any of the specified covariance structures.

In conclusion, test statistics T_1 and T_2 are unable to control the type I error rate according to Bardley's criterion and are therefore not suitable for analyzing this type of data. In contrast, T_3 , T_4 and T_5 are capable of controlling the type I error rate in accordance with Bardley's (1978), and the results are consistent for both $k = 3$ and $k = 4$.

From Fig. 1 - Fig. 10, considering the test statistics under the criterion of Bardley's (1978) and the specified covariance structures, it is found that T_3 exhibits the highest empirical power in several structures, namely the First-Order Autoregressive (AR1), Simple, and Variance Components structures. The second highest power is observed for T_5 and T_4 .

For the Compound Symmetry and Banded Toeplitz structures, T_4 demonstrates the highest empirical power, followed by T_3 and T_5 , respectively. For all three statistics, the power values approach 1 as the sample size and the number of variables increase.

In conclusion, T_3 shows the highest overall power compared with T_4 and T_5 , and the results are consistent for both population sizes, $k = 3$ and $k = 4$.

Application to a Real Dataset for Evaluating Test Performance

In addition, the proposed methods were applied to a glioma gene expression dataset. Variables were selected based on their coefficients of variation restricted to the interval [0.25–0.5]. To evaluate testing performance on real data, the dataset consists of $k = 4$ groups with $p = 24,785$ variables and a total sample size of $n = 180$. The group sample sizes are $n_1 = 26$, $n_2 = 81$, $n_3 = 50$ and $n_4 = 23$, and the results are reported in Table 1.

Table 1 Test statistics and p-values of the statistical tests T_3 , T_4 and T_5 in the gene expression dataset of glioma brain tumors

	T_3	T_4	T_5
Test Stat	2.0469	5.9589	29.3143
p-value	0.0203	0.1136	< 0.001

According to the results presented in Table 1, the test statistics T_3 and T_5 , which follow a normal distribution, yielded test statistic values of 2.0469 and 29.3143, with corresponding p-values of 0.0203 and < 0.001 , respectively. Since both p-values are less than the significance level of 0.05, the null hypothesis was rejected, indicating that the covariance matrices of gene expression in glioma tumors differ among at least two groups at the 0.05 significance level.

In contrast, for the test statistic T_4 , which follows a chi-square distribution, the test statistic value was 5.9589 with a p-value of 0.1136. As this p-value exceeds the 0.05 significance level, the null hypothesis was accepted.

From the analysis of the real dataset after filtering variables based on coefficients of variation within the interval $[0.25 - 0.5]$, it was found that test statistics T_3 and T_5 were able to detect differences in the covariance matrices of the real data. Considering power, T_3 exhibited the highest power in several scenarios. This suggests that T_3 is the most efficient method for data analysis, providing the most reliable testing results among the considered statistics.

Conclusion

From the comparative study of the performance of five test statistics for testing the equality of covariance matrices across multiple populations with high-dimensional data, T_1 by Schott (2007), the test statistic T_2 by Srivastava and Yanagihara (2010) the test statistic T_3 by Ahmad (2017) the test statistic T_4 by Qayed and Han (2020), and the test statistic T_5 by Sun et al. (2022), the results are consistent for both population sizes, $k = 3$ and $k = 4$.

The findings indicate that T_3 , T_4 and T_5 are able to control the type I error rate within the acceptable range of Bardley (1978), i.e., $[0.025 - 0.075]$, under all specified covariance structures, particularly as the sample size and the number of variables increase. Among these, T_4 provides type I error rates closest to the nominal significance level of 0.05 across all covariance structures when compared with the other methods.

In contrast, T_1 and T_2 are not suitable for high-dimensional data analysis because they fail to control the type I error rate within Bardley's (1978) criterion under all specified covariance structures. Specifically, T_1 is unable to control the type I error rate under any covariance structure considered, while T_2 satisfies the criterion under only two structures: Compound Symmetry and Simple.

When considering the empirical power of the tests that satisfy Bardley's criterion, T_3 demonstrates the highest power under the First-Order Autoregressive (AR1), Simple, and Variance Components structures, followed by T_5 and T_4 , respectively. Under the Compound Symmetry and Banded Toeplitz structures, T_4 shows the highest power, followed by T_3 and T_5 . The power of T_3 , T_4 and T_5 approaches 1 as the sample size and the number of variables increase.

Although T_4 provides type I error rates closest to the nominal level of 0.05, its power is lower than that of T_3 and T_5 under the First-Order Autoregressive (AR1), Simple, and Variance Components structures. Consequently, T_3 exhibits the highest overall power among the considered methods.

When applying the findings to real data and selecting a test statistic that is both efficient and maintains the type I error rate within [0.025 - 0.075], T_3 is capable of detecting differences in covariance matrices when they exist. In empirical data analysis involving two groups with a significance level of 0.05, and considering both type I error control and power, T_3 yields the most reliable results.

In summary, T_3 is the most efficient and reliable test statistic in this study due to its ability to control the type I error rate, its high power, and its effectiveness in detecting differences in real datasets.

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